

Proper Completion in Workflow Nets

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Solution

Let $N = \langle \mathcal{P}, \mathcal{T}, \mathcal{F} \rangle$ be a proper workflow net with the initial marking M_0 . We proceed by contradiction.

1. Suppose there exists a marking $M_{bad} \in \mathcal{R}(N, M_0)$ such that $M_{bad}(\text{out}) = 1$, and there is at least one other place $p_k \in \mathcal{P} \setminus \{\text{out}\}$ for which $M_{bad}(p_k) > 0$. Let X represent the non-empty multiset of tokens in $\mathcal{P} \setminus \{\text{out}\}$ under M_{bad} .
2. Since M_{bad} is reachable from M_0 , the final marking M_{out} must be reachable from M_{bad} . Therefore, there exists a firing sequence σ such that $M_{bad} \xrightarrow{\sigma} M_{end}$.
3. The place 'out' has no outgoing arcs. Consequently, the token in 'out' under marking M_{bad} cannot be consumed by any transition in σ . For M_{bad} to transition to M_{end} , the sequence σ must:
 - Not add any further tokens to 'out'.
 - Remove all tokens from the multiset X .
4. For the tokens in X to be entirely removed by σ without any of them being added to 'out', they must be eventually consumed by some transition that doesn't produce any tokens.
5. Using the definition of a WF-net, every node must be on some path from 'in' to 'out'. Existence of a transition that doesn't produce any tokens directly contradicts that. Thus, the initial assumption must be false. It follows that if $M \in \mathcal{R}(N, M_0)$ and $M(\text{out}) = 1$, then M must be equal to M_{end} .